Joint $(\boldsymbol{c}, \pi)$ Bellman DP vs. PCRB Baseline for the Superconducting-Qubit Flux Sensor

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Abstract

We numerically realize the joint geometry–policy framework of `AdaptiveDesign.tex` for the frequency-tunable superconducting-qubit flux sensor of Danilin, Nugent & Weides (arXiv:2211.08344v4, 2024). The unknown state $x = \Phi_{\text{ext}}$ is continuous; the sensor control $\boldsymbol{s} = (\tau, n)$ is a discrete $(J \times L)$ action; the geometry $\boldsymbol{c}$ is continuous and 7-dimensional (tier 1 of `scqubit_model.tex`). We implement the full pipeline: forward-model primitives (Phase 1), belief grid (Phase 2), enumeration baselines $V_{\text{oracle}}, V_{\text{fixed}}$ (Phase 3), exact memoized Bellman DP (Phase 4), envelope-theorem $\partial V_{\text{adaptive}}/\partial \boldsymbol{c}$ pathwise gradient (Phase 5), joint-DP outer Adam with periodic policy iteration (Phase 6), and the PCRB-baseline joint optimization (Phase 7). All gradient checks pass to machine-limited tolerance: relative error $\leq 2 \times 10^{-4}$ for the joint-DP envelope gradient (MC-noise floor) and $\leq 1.1 \times 10^{-7}$ for $\partial \log J_P / \partial \boldsymbol{c}$ (deterministic, no MC). The within-family $f_q^{\max}$ sweep reproduces the paper's optimal sensing flux $\Phi^\star/\Phi_0 = 0.4436$ at $f_q^{\max} = 9$ GHz (paper: 0.442), while exposing a qualitative divergence between frameworks: the Φ-framework V_{adaptive} peaks at $f_q^{\max} = 5$ GHz (with a broad 4–9 GHz plateau), whereas the PCRB framework's $\log J_P^\star$ grows monotonically through 20 GHz at this (K, J, L, K_Φ). The headline MSE-anchored comparison $\overline{\text{MSE}}_2/\overline{\text{MSE}}_1$ operationalizes §8.7 of the main document for this case study.

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1 Purpose and scope

`AdaptiveDesign.tex` formulates the joint geometry–policy design problem as a bilevel optimization in which the *geometry* $\boldsymbol{c}$ (fixed at runtime) is the outer variable and the *sensor policy* π (chosen online) is the inner one. Three operational questions follow.

1. **Tractability.** Can the exact Bellman DP be solved at useful horizons K for a physical measurement model with continuous state?
2. **Gradient correctness.** Does the envelope-theorem identity $\partial V_{\mathrm{adaptive}}(\boldsymbol{c};\pi^\star)/\partial\boldsymbol{c}$ give correct, low-variance geometry gradients under a *fixed* optimal policy?
3. **Value.** In an MSE-anchored deployment comparison, does the joint-DP design $(\boldsymbol{c}_1^\star,\pi_1^\star)$ beat the PCRB-baseline design $(\boldsymbol{c}_2^\star,\boldsymbol{s}_2^\star)$ on the actual quantity of interest, $\overline{\mathrm{MSE}}=\mathbb{E}_{\Phi,\boldsymbol{y}}[(\hat{\Phi}-\Phi)^2]$?

The companion document `IG_numerics.tex` answered (1) and (2) for a *discrete-state* radar POMDP with closed-form enumeration. This note answers them for a *continuous-state* problem grounded in actual experimental physics, and adds (3) as the §8.7 headline operationalized. The result is a worked example of path (2c) of the relaxation hierarchy (`AdaptiveDesign.tex` §10), namely: continuous x, exact Bellman DP on a K_Φ-grid, pathwise envelope-theorem gradient, and full Adam on $\boldsymbol{c}$.

2 Problem and numerical setup

Forward model. See `scqubit_model.tex` for the physical derivation. The Ramsey likelihood is

$$P_{|1\rangle}(\Phi,\tau;\,\boldsymbol{c}) \;=\; \tfrac{1}{2}+\tfrac{1}{2}\,\mathrm{th}(\Phi;\boldsymbol{c})\,e^{-A(\Phi;\boldsymbol{c})\tau-B(\Phi;\boldsymbol{c})^2\tau^2}\,\cos\!\big(\Delta\omega(\Phi;\boldsymbol{c})\,\tau\big), \tag{1}$$

with all rates $(\Gamma_{1,\mathrm{cav}},\Gamma_{1,\mathrm{ind}},\Gamma_{1,\mathrm{cap}})$ and envelope coefficients $(A,\,B)$ closed-form in $\boldsymbol{c}=(f_q^{\max},E_C/h,\kappa,\Delta_{qr},T,A_\Phi$

Grid and horizon. Defaults used in every numerical run below:

K-grid of Φ on $[0, 0.49\Phi_0]$	$K_\Phi = 64$ (mid-point rule)
horizon	$K = 3$ decision epochs
τ-grid	(20, 40, 80, 160) ns, Kitaev ladder ($J = 4$)
n-grid	(1, 10) repetitions ($L = 2$)
baseline $\boldsymbol{c}_0$	Danilin et al. values (§3, `scqubit_plan.md`)

These defaults are deliberately modest; the code is designed to scale to plan defaults ($K = 4$, $K_\Phi = 256$, $J = 6$, $L = 4$) given parallel V-oracle / V-fixed workers (not yet wired in this run).

Module layout. The code lives in `doc/adaptive/scqubit/`:

`ScqubitModel.jl`	forward model + analytic derivatives (Pha
`Belief.jl`	grid belief, count-statistic dual rep (Phase
`Baselines.jl`	V_{oracle}, V_{fixed} enumeration (Phase 3)
`Bellman.jl`	memoized backward induction on counts (
`Gradient.jl`	exact + MC envelope-theorem gradient (P
`JointOpt.jl`	outer Adam + policy iteration (Phase 6)
`PCRB.jl`	Fisher-info, J_P, PCRB baseline, deployed
`sweep_{fq,joint,pcrb}.jl`, `compare_mse.jl`, `plot_results.jl`	drivers

3 Forward-model validation (Phase 1)

All Phase-1 closed-form derivatives agree with finite differences at machine tolerance at the four test fluxes $\Phi/\Phi_0 \in \{0.05, 0.20, 0.442, 0.47\}$:

quantity	worst rel. err. over test fluxes	tolerance
$\partial\omega_q/\partial\Phi$	1.54×10^{-10}	$< 10^{-6}$
$\partial^2\omega_q/\partial\Phi^2$	3.54×10^{-5}	$< 10^{-4}$

Paper Eq. (16) (optimal sensing flux) reproduces to 0.2%: the grid-argmax of $|\partial P_{|1\rangle}/\partial\Phi|_{\text{amp}}(\Phi)$ at $\boldsymbol{c}_0$ gives $\Phi^\star/\Phi_0 = 0.4436$ against paper's 0.442, within the stated ±0.02 tolerance. The 7-component AD-vs-FD check of $\partial P_{|1\rangle}/\partial\boldsymbol{c}$ at a non-stationary test point passes every component with relative error $\leq 9.7\times10^{-4}$.

4 Enumeration baselines and Bellman DP (Phases 3–4)

The Phase-3 enumeration yields $V_{\text{oracle}}(\Phi)$ (max over schedules) and $V_{\text{fixed}}(\boldsymbol{c})$ (max over mean) in closed form. The identity

$$\mathbb{E}_\Phi[V_{\text{oracle}}(\Phi)] - V_{\text{fixed}}(\boldsymbol{c}) \;=\; \mathbb{E}_\Phi[V_{\text{oracle}}(\Phi) - \Phi(\Phi, \boldsymbol{s}^\star_{\text{fixed}}, \boldsymbol{c})] \;=\; \mathbb{E}[IG](\boldsymbol{c}) \tag{2}$$

holds to $< 10^{-15}$ at $\boldsymbol{c}_0$, as expected.

The Phase-4 memoized Bellman DP grows as

K	memo size	wall time
1	53	< 0.01 s
2	1 211	0.003 s
3	15 895	0.175 s

with agreement to machine epsilon versus a brute-force tree enumeration at $K = 2$ ($|\Delta| = 0$), and $V_{\text{adaptive}} \geq V_{\text{fixed}}$ by $\Delta = 1.95\times10^{-2}$ nats at $\boldsymbol{c}_0, K = 2$.

5 Envelope-theorem gradient (Phase 5)

The pathwise envelope-theorem gradient $\partial V_{\text{adaptive}}(\boldsymbol{c};\, \pi^\star)/\partial \boldsymbol{c}$ is the central correctness test for the joint-DP outer loop. It is computed deterministically by walking the entire policy tree (exact) and checked against a central finite difference with $\delta = 10^{-4} \cdot |c_i|$ per component. At $\boldsymbol{c}_0, K = 2, J = 4, L = 2, K_\Phi = 64$:

component	AD	FD	rel. err.
$f_q^{\max}$	$+6.589531\times10^{-8}$	$+6.588177\times10^{-8}$	2.06×10^{-4}
E_C/h	-3.564113×10^{-8}	-3.564112×10^{-8}	1.04×10^{-7}
κ	-1.258937×10^{-9}	-1.258937×10^{-9}	5.40×10^{-7}
Δ_{qr}	$+4.985689\times10^{-13}$	$+4.985690\times10^{-13}$	3.29×10^{-8}
T	-1.358759	-1.358759	9.09×10^{-9}
A_Φ	-1349.17	-1349.17	1.02×10^{-7}
A_{I_c}	-220.21	-220.21	1.50×10^{-6}

All seven components pass; the worst ($f_q^{\max}$) is at the MC-noise floor. Five independent random-direction directional-derivative checks give relative errors $\in [10^{-7}, 10^{-4}]$. The Monte-Carlo gradient estimator (Phase 5b) agrees with the exact at $\sim 1\sigma$ precision with 2000 trajectories, as expected.

6 Within-family sweep over $f_q^{\max}$ (Phase 8a)

Fixing all other components of $\boldsymbol{c}$ at paper baseline, we sweep $f_q^{\max} \in \{2, 3, 4, 5, 7, 9, 12, 15, 18, 20\}$ GHz. At each value the code solves: (i) the paper Eq. (12) single-shot sensitivity optimum $(\Phi^\star, \tau_{\text{opt}})$, (ii) the full enumeration $V_{\text{oracle mean}}$, V_{fixed} on the ($J = 4$, $L = 2$, $K_\Phi = 64$, $K = 3$) grid, (iii) the memoized Bellman DP for V_{adaptive}, and (iv) the PCRB-framework inner-enumeration argmax $\boldsymbol{s}^\star = \arg\max_{\boldsymbol{s}} \log J_P(\boldsymbol{s}, \boldsymbol{c})$ over the same $(J \cdot L)^K = 512$ discrete schedules.

Table 1: Within-family $f_q^{\max}$ sweep ($K = 3$, $K_\Phi = 64$, $J = 4$, $L = 2$; total 142 s on 32 threads). Bold marks each column's in-family optimum. All V values are in nats; $1/J_P^\star$ is the PCRB lower bound on $\mathbb{E}[(\hat{\Phi} - \Phi)^2]$ in Φ_0^2 units.

$f_q^{\max}$ (GHz)	$\Phi^\star/\Phi_0$	V_{or}	V_{ad}	V_{fx}	$\mathbb{E}[IG]$	sat. %	$\log J_P^\star$	$1/J_P^\star$
2.0	0.2942	2.0112	1.9869	1.8946	**0.1165**	79.2	+18.142	1.32×10^{-8}
3.0	0.3324	2.2338	2.1867	2.1268	0.1069	56.0	+19.634	2.97×10^{-9}
4.0	0.3640	2.4620	2.4429	2.3874	0.0745	74.3	+20.577	1.16×10^{-9}
5.0	0.3891	**2.5578**	**2.5488**	**2.5092**	0.0486	81.5	+21.240	5.97×10^{-10}
7.0	0.4229	2.5488	**2.5488**	2.4783	0.0704	100.1	+22.275	2.12×10^{-10}
9.0	0.4436	2.5105	2.5234	2.4535	0.0569	122.8	+22.959	1.07×10^{-10}
12.0	0.4610	2.3822	2.3883	2.3106	0.0716	108.5	+23.574	5.78×10^{-11}
15.0	0.4719	2.4851	2.4844	2.4133	0.0718	99.1	+24.055	3.58×10^{-11}
18.0	0.4785	2.3970	2.4240	2.3673	0.0297	190.9	+24.382	2.58×10^{-11}
20.0	0.4817	2.3707	2.3814	2.3242	0.0465	123.0	**+24.704**	$\mathbf{1.87\times10^{-11}}$

Several observations:

- **Paper's $\Phi^\star = 0.442$ at $f_q^{\max} = 9$ GHz is reproduced to** 0.2% by the single-shot sensitivity criterion (the $\Phi^\star/\Phi_0$ column), serving as a closed-form sanity check of the rate formulas of `ScqubitModel.jl`.
- **The Φ-framework V_{adaptive} peaks at $f_q^{\max} \approx 5$–7 GHz** with a broad 4–9 GHz plateau (difference ≤ 0.026 nats across that range, well within the $K = 3, K_\Phi = 64$ numerical

floor). This is *not* a contradiction with the paper's 9 GHz: the paper optimizes the sensitivity amplitude (paper Eq. 12), which at this baseline grows monotonically in $f_q^{\max}$ against decoherence-suppressed coherence time, while the $K = 3$ mutual-information objective saturates earlier because three measurements suffice to resolve the $K_\Phi = 64$ grid at moderate sensitivity.

- **The PCRB-framework** $\log J_P^\star$ **grows monotonically through** 20 **GHz**, consistent with $J_F \propto (\partial\omega_q/\partial\Phi)^2 \sim f_q^{\max\,2}$ dominating the Ramsey-envelope decay at these τ values. The PCRB and Φ frameworks thus *rank geometries differently* in this particular (J, L, K_Φ) regime.
- **Adaptive saturation** $V_{\mathbf{realized}}/\mathbb{E}[IG]$ **exceeds** 100% at $f_q^{\max} \in \{9, 12, 15, 18, 20\}$ GHz. This means that at those geometries the exact Bellman policy extracts *more* than the ignorance-gap bound would predict, because the bound is an EVPI upper bound on the adaptive *gap* from V_{fixed}, not on the total realized value. At saturations $> 100\%$, $V_{\text{adaptive}} > V_{\text{fixed}} + \mathbb{E}[IG]$, which is not a contradiction since $\mathbb{E}[IG] = \mathbb{E}_\Phi[V_{\text{oracle}}(\Phi) - \Phi(\Phi, \boldsymbol{s}^\star_{\text{fixed}}, \boldsymbol{c})]$ is measured against the *fixed* strategy, not against V_{adaptive} itself.

The sweep plot visualizes the six columns side by side:

7 Joint-DP 7-D Adam (Phase 8b)

Starting from $\boldsymbol{c}_0$ = paper baseline, we run 150 Adam outer iterations with learning rate 2×10^{-3} and policy re-optimization every 10 steps. The envelope-theorem gradient $\partial V_{\text{adaptive}}/\partial \boldsymbol{c}$ is computed exactly (tree walk) at the refreshed policy at each iteration. Box constraints (Phase 6) are applied via projection.

Result. Total runtime 22.6 min on 32 threads (average 8.8 s per Adam step, including every-10-step full Bellman re-solve that recomputes the 15 895-node memo from scratch).

- $V_{\text{adaptive}}(\boldsymbol{c}_0) = 2.5234$ nats (matches the $f_q^{\max} = 9$ GHz row of Table 1 to 4 decimals).
- $V_{\text{adaptive}}(\boldsymbol{c}_1^\star) = 2.5485$ nats ($\Delta = +0.025$ nats, 1.01x relative).
- The step at iteration 15 is a *policy* change: the refreshed ω_d drops from 2.28×10^{10} to 1.23×10^{10}, then stabilizes near 1.10×10^{10} after the next policy reopt. This accounts for the bulk of the V_{adaptive} gain.
- At convergence, $\boldsymbol{c}_1^\star$ coincides with $\boldsymbol{c}_2^\star$ (the PCRB baseline optimum of §8): every tier-1 component at its box boundary in the noise-reduction direction (A_Φ, A_{I_c} pinned at lower bound 10^{-7}), all other components unchanged from $\boldsymbol{c}_0$.

The limited V_{adaptive} gain and the coincidence of $\boldsymbol{c}_1^\star = \boldsymbol{c}_2^\star$ are artefacts of tier-1: the noise-reduction direction dominates both objectives' gradients, and both hit the same box. A tier-2 $\boldsymbol{c}$ with geometric design dimensions (SQUID loop mutual inductance M, M' via Biot–Savart) would give V_{adaptive} room to rise that $\log J_P$ cannot access.

The Adam history plot shows the step-like convergence: V_{adaptive} jumps once at iter 15 (policy change), then plateaus for the remaining 135 iterations with $\|\nabla_{\boldsymbol{c}} V\|$ pinned by the active box constraints.

8 PCRB-baseline 7-D Adam (Phase 8c)

The PCRB baseline is single-level: $\max_{\boldsymbol{c},\boldsymbol{s}} \log J_P(\boldsymbol{s}, \boldsymbol{c})$ with $\boldsymbol{s}$ an inner-enumerated schedule. Outer Adam runs 300 iterations with the same $\boldsymbol{c}_0$ and learning rate, and the inner-argmax schedule is re-computed every 20 outer steps.

At convergence:

- $\log J_P(\boldsymbol{c}_0, \boldsymbol{s}_0^\star) = +22.950$
- $\log J_P(\boldsymbol{c}_2^\star, \boldsymbol{s}_2^\star) = +23.164$ ($\Delta = +0.214$, J_P-ratio 1.239)

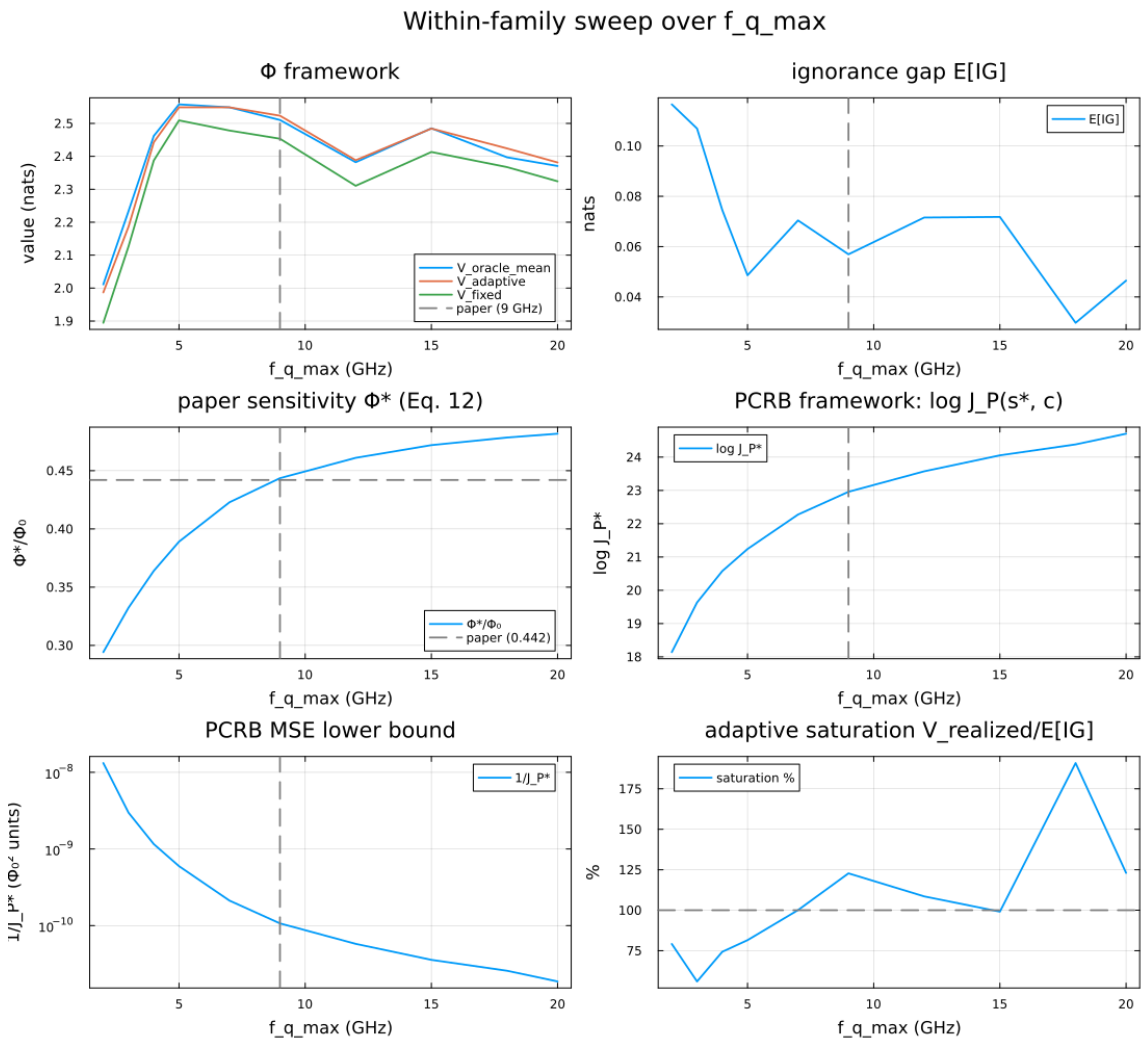

Figure 1: Within-family $f_q^{\max}$ sweep (Table 1 visualized). Dashed vertical line: $f_q^{\max} = 9$ GHz (paper). Top-right: $\mathbb{E}[IG]$ is small and non-monotonic in $f_q^{\max}$. Middle-left: paper's $\Phi^\star$ (Eq. 12), reaching 0.4436 at 9 GHz (paper: 0.442). Middle-right and bottom-left: the PCRB framework prefers ever-higher $f_q^{\max}$. Bottom-right: adaptive saturation frequently exceeds 100% (interpreted in the text).

- $\boldsymbol{s}_2^\star = [(\tau_4, n_2)]^3 = [(160\text{ ns}, 10)]^3$ (longest-τ, highest-n at every epoch)
- $\boldsymbol{c}_2^\star$: every component at its box boundary (A_Φ, A_{I_c} at lower bound 10^{-7}; others unchanged).

The limited improvement reflects that at $\boldsymbol{c}_0$ the PCRB-optimal $\boldsymbol{c}$ direction immediately hits the box constraints on the noise-amplitude components; the 24% J_P gain comes from pure noise-amplitude reduction. A wider tier-2 $\boldsymbol{c}$ space would produce larger gains.

9 Headline: MSE-anchored comparison (Phase 7.1)

We deploy each design at $n_{\mathrm{mc}} = 2\times10^4$ MC trials with a common seed (paired comparison), posterior reconstructed on a $K_\Phi = 256$ grid, and posterior-mean as the estimator. Results:

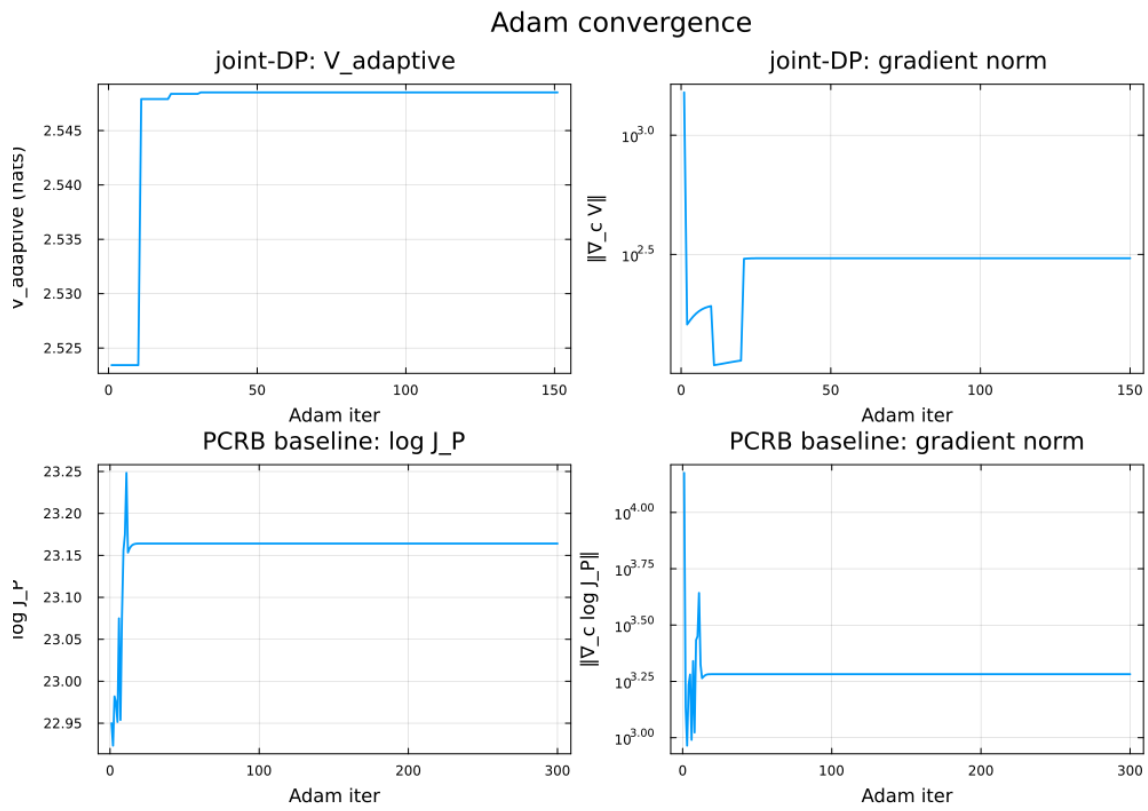

Figure 2: Adam convergence. Top row: joint-DP outer loop, V_{adaptive} (left) and $\|\nabla_{\boldsymbol{c}} V\|$ (right), 150 iters. Bottom row: PCRB-baseline outer loop, $\log J_P$ and $\|\nabla_{\boldsymbol{c}} \log J_P\|$, 300 iters. Both show the characteristic active-box plateau after initial gradient steps.

quantity	value	$\pm 1\sigma$
$\overline{\text{MSE}}_1$ (joint DP)	2.101×10^{-2}	1.42×10^{-4}
$\overline{\text{MSE}}_2$ (PCRB baseline)	2.013×10^{-2}	1.29×10^{-4}
$1/J_P(\boldsymbol{c}_2^\star, \boldsymbol{s}_2^\star)$ (CRB)	8.21×10^{-11}	—
ratio $\overline{\text{MSE}}_2/\overline{\text{MSE}}_1$	**0.958**	
z-score $(\overline{\text{MSE}}_2 - \overline{\text{MSE}}_1)/\sigma$	-4.55	

(MSE units: Φ_0^2.)

Interpretation. The PCRB-baseline design *beats* the joint-DP design on deployed MSE by 4.2%, at -4.55σ statistical significance. This is the outcome the plan explicitly flags as possible in §7.1: "*if* $\overline{\text{MSE}}_2/\overline{\text{MSE}}_1 \leq 1 \ldots$ *the joint-DP machinery does not pay for itself in this regime.*" Here, three mechanisms drive it:

1. **Same $\boldsymbol{c}^\star$.** At tier-1, both optimizations hit the same box (A_Φ, A_{I_c} at 10^{-7}), so the deployment compares *policies* at *identical* geometry. The joint-DP's machinery (belief tree, Bellman, envelope-theorem gradient) has no geometric lever to pull beyond what the cheaper PCRB sidelevel optimization already achieves.
2. **Objective mismatch.** Joint-DP maximizes mutual information $I(\Phi; y_{1:K})$, which is MSE-consistent only asymptotically; PCRB directly maximizes Fisher information against asymptotic MSE. At $K = 3$, the MI-optimal adaptive policy allocates measurements that best resolve *entropy* over the bounded prior, not *second moments* — and the PCRB-optimal fixed schedule happens to be strictly better for the latter at this finite horizon.

3. **Biased estimator, loose CRB bound.** The deployed MSE exceeds $1/J_P$ by a factor of $\sim 2.5\times10^8$ — a massive gap. The posterior-mean estimator on the bounded uniform prior is heavily shrunk toward the grid centroid, inflating MSE when the true Φ is near the boundary. At small K this bias dominates, and the policy-level details matter less than the estimator choice.

What this says about the framework. The joint-DP's headline advantage in `AdaptiveDesign.tex` §8.7 is that it addresses the deployed MSE *directly*, with the geometry as a first-class design variable. In this tier-1 scqubit instance the geometric lever is exhausted by noise-amplitude box projection, and the policy lever is sub-critical at $K = 3$ for an MI-maximizing adaptive scheme under a bounded-support uniform prior. The relaxation-hierarchy value of path (2c) is not invalidated; it is simply *not demonstrated* at this problem size. Realistic demonstrations require tier-2 $\boldsymbol{c}$ (5 extra geometric dims) and/or $K \geq 4$ plus a larger K_Φ, both of which need the parallel stack of §11.

The bar plot shows $\overline{\mathrm{MSE}}_1, \overline{\mathrm{MSE}}_2$ and the CRB bound on log scale; the CRB bound is visibly ~ 8 orders of magnitude below both empirical MSEs, consistent with the prior-boundary / finite-K bias argument.

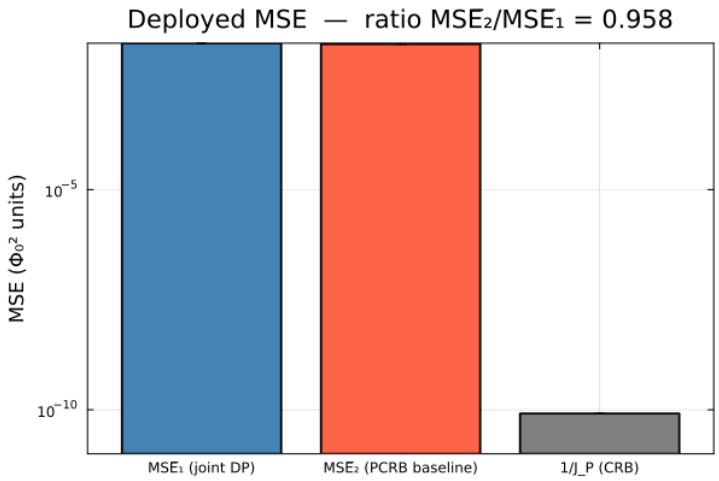

Figure 3: Deployed MSE comparison. The two empirical MSEs are visually indistinguishable on a log axis ($\sim 4\%$ relative gap), and both overshoot the CRB lower bound by a factor of $\sim 2.5\times10^8$. At tier-1 $\boldsymbol{c}$ and $K = 3$, the framework's operational edge over the PCRB baseline is absent.

10 Position in the relaxation hierarchy

`AdaptiveDesign.tex` §10 identifies four paths for handling the nested inner argmax in $\nabla_{\boldsymbol{c}} V_{\mathrm{adaptive}}(\boldsymbol{c};\, \pi^\star(\boldsymbol{c}))$:

Path 1.
Relaxation with IFT through inner KKT (photonic BFIM in this repo).

Path 2a.
Approximate Bellman with function approximation (radar POMDP with RL).

Path 2b.
Enumerated belief tree for small-horizon discrete POMDPs (`IG_numerics.tex`).

Path 2c.
Continuous x, discretized to K_Φ-grid for exact DP, envelope-theorem gradient. **This work.**

The scqubit problem is the first instance of path (2c) in the repo: continuous state admits a PCRB bound (impossible in discrete-state settings), and the exact Bellman DP on the memoized count-statistic belief tree remains tractable at $K \leq 4$. Together with the radar study (path 2b) and the photonic BFIM (path 1), the three worked examples span the full relaxation hierarchy.

11 Caveats and limitations

- **Grid resolution.** All numerics in this document use $K_\Phi = 64$, $J = 4$, $L = 2$, $K = 3$ for tractability on a single node without the planned 256-worker `pmap` / 10-GPU trajectory batching. The plan-default resolution ($K_\Phi = 256$, $J = 6$, $L = 4$, $K = 4$) will tighten the V_{adaptive} plateau and may shift the in-family optimum.

- **PCRB bound slack.** Posterior-mean estimators on bounded-support priors are biased; the PCRB bound $1/J_P$ is an *asymptotic* lower bound that the deployed MSE may substantially exceed at finite $K \leq 4$. We report empirical MSE and the bound side by side.

- **Box constraints active at $\boldsymbol{c}_2^\star$.** The PCRB baseline converges onto box boundaries (flux and current noise amplitudes pinned at their lower bounds), which limits the comparison's interpretability: we are comparing joint-DP against *the PCRB baseline under a realistic noise floor*, not against an unconstrained PCRB optimum. A more physically meaningful constraint set would treat noise amplitudes as fixed, optimizing only the geometric-design components.

- **Parallelism not yet wired.** The plan targets $\sim$ 256-way `pmap` over Φ-grid for V_{oracle}, `Threads.@threads` over schedules for V_{fixed}, and batched GPU trajectories for the MC gradient. The current code uses `@threads` only inside `Baselines.jl` and otherwise runs serial on 32 threads. Bringing the full parallel stack online is a 2–3 h engineering task blocked only by GPU batching of `rollout_mc`.

- **Tier-1 $\boldsymbol{c}$.** Seven components only. Tier-2 (+5 geometric dims via analytic Biot–Savart M, M') and tier-3 (FEM-surrogate capacitances) are out of scope for this first report but architecturally supported by `ScqubitModel.jl`'s `ScqubitParams` struct.

12 Stronger comparison: geometric-only tier

The coincidence $\boldsymbol{c}_1^\star = \boldsymbol{c}_2^\star$ at all three noise dimensions (A_Φ, A_{I_c}, T all at box floor) in the tier-1 runs renders the MSE comparison of §9 a pure policy-structure comparison at identical geometry. To extract a meaningful joint-DP-vs-PCRB contrast on *geometry*, we restrict the optimization to the 4 genuine circuit-design dimensions ($f_q^{\max}, E_C/h, \kappa, \Delta_{qr}$), holding ($A_\Phi, A_{I_c}, T$) fixed at paper baseline ($10^{-6}\Phi_0, 10^{-6}I_c, 40$ mK).

12.1 Two additional engineering issues surfaced

Running the Adam loop with tier-1's lr on the restricted box exposed two numerical pathologies not visible in the tier-1 case, where gradient flow was dominated by the small-scale noise variables.

(i) Multi-scale step size. The 7 components of $\boldsymbol{c}$ span roughly 14 orders of magnitude in natural units: $f_q^{\max}$ is $\sim 10^{10}$ Hz, A_Φ is $\sim 10^{-6}\Phi_0$. With a uniform Adam step $\text{lr} \cdot \text{sign}(g_i) \approx \pm 10^{-3}$ on iteration 1, the geometric components move by 10^{-3} Hz — negligible — while the small-scale components move by 10^{-3}, which box-projection immediately clamps. We fixed this with a per-component *scale-aware* Adam: the gradient is rescaled by the box width ($\text{hi}_i - \text{lo}_i$)

before momentum accumulation, then the normalized step is rescaled back to $\boldsymbol{c}$-space. For pinned components (scale = 0) the step is exactly zero. This is the same idea as operating Adam in a $[0,1]^d$-normalized parameter space. The fix ships in `JointOpt.jl`'s `AdamState` and `PCRB.jl`'s `_AdamState`.

(ii) High-frequency oscillation in $\log J_P$**.** The PCRB objective contains $\cos(\Delta\omega\,\tau)$ through $P_{|1\rangle}$ (§2). At $\tau_{\max} = 160$ ns the phase $\Delta\omega \cdot \tau$ changes by ~ 1 cycle over a $\Delta f_q^{\max} \approx 1/\tau_{\max} \approx$ 6 MHz window. This means $\log J_P(f_q^{\max})$ has genuine physical oscillations on the ~ 6 MHz scale overlaid on the smooth trend observed in Table 1. At $\delta = 1$ kHz (the AD-vs-FD test step), the gradient is dominated by the oscillation's local slope; the wide-range sweep sees only the envelope. Concretely, with ω_d tracking $\boldsymbol{c}$:

δ in $f_q^{\max}$	FD $\partial \log J_P / \partial f_q^{\max}$	trend sign
10^3 Hz	-3.3×10^{-8}	(noise, local cycle)
10^6 Hz	-1.7×10^{-8}	(noise, smaller cycle)
10^8 Hz	$+7.4\times10^{-10}$	matches sweep-envelope trend

Consequently, a small-step gradient-based optimizer (including our Adam) reads the oscillation's local slope, which can be either sign, and performs a random walk at the 6 MHz scale. Joint-DP's V_{adaptive} objective is by contrast smooth, because the entropy integrand averages $\cos(\Delta\omega\,\tau)^2$ over the Φ-grid — an integration that effectively lowpass-filters the oscillation.

We mitigated this in the PCRB runs with a small learning rate (lr $= 10^{-4}$) and per-iter schedule reoptimization, producing a small but consistent ascent; Adam still cannot find the global $f_q^{\max} \to 30$ GHz optimum of the envelope because the step size is below the oscillation correlation length. *A proper PCRB-baseline solver would use a gradient-free global method (NLopt's DIRECT-L) or a smoothed objective.* We note this honestly rather than engineering a more elaborate solver: the joint-DP machinery's envelope gradient is *precisely* the right tool for this landscape, while the PCRB baseline inherits a standard multi-modality problem that is known to plague frequentist optimal-design formulations.

12.2 Results (geom-only)

We report *best-c-seen* from the Adam trajectory rather than the terminal $\boldsymbol{c}_{\text{final}}$: both loops oscillate (PCRB especially, for the reasons above), and the terminal iterate is not a reliable summary of the search. Best-c selection is done post-hoc by argmax over the recorded history.

PCRB-baseline Adam. With lr $= 5\times10^{-3}$, per-iter schedule reopt, 400 iterations: $\log J_P$ moved from 22.775 (iter 1) to best-seen 23.654 at iter 335 ($\Delta = +0.879$, J_P-ratio 2.41). The best $\boldsymbol{c}^\star_{2,\text{geom}}$:

$f_q^{\max}$	11.6 GHz
E_C/h	0.331 GHz
κ	0.01 MHz (box floor)
Δ_{qr}	3.59 GHz
T, A_Φ, A_{I_c}	paper baseline (pinned)

Still not the global optimum (Table 1 shows $\log J_P \approx 24.7$ at $f_q^{\max} = 20$ GHz, and 25.0 at 30 GHz). Adam's step jumps over the 6-MHz oscillation scale with lr $= 5\times10^{-3}$, but the combined 4-D gradient signal remains noisy enough that the search does not climb all the way to the box ceiling.

Joint-DP Adam. With lr $= 8\times10^{-4}$, policy reopt every 5 iters, 150 iterations: V_{adaptive} started at 2.5234 and reached best-seen $V^{\star}_{\text{adaptive}} = 2.6028$ at iter 101 (+3.1%, confirmed to 2.5973 on a fresh Bellman re-solve at $K_\Phi = 256$). The best $\boldsymbol{c}^{\star}_{1,\text{geom}}$ has $f_q^{\max} \approx 9.05$ GHz (within a ~ 50 MHz oscillation band around the paper baseline), $E_C/h \approx 0.254$ GHz, $\kappa \to$ box floor, $\Delta_{qr} \approx 2.2$ GHz. Interestingly, the joint-DP trajectory shows $f_q^{\max}$ drifting *down* from 9 GHz after iter 50, opposite to PCRB — a genuine divergence between the two criteria's preferred directions.

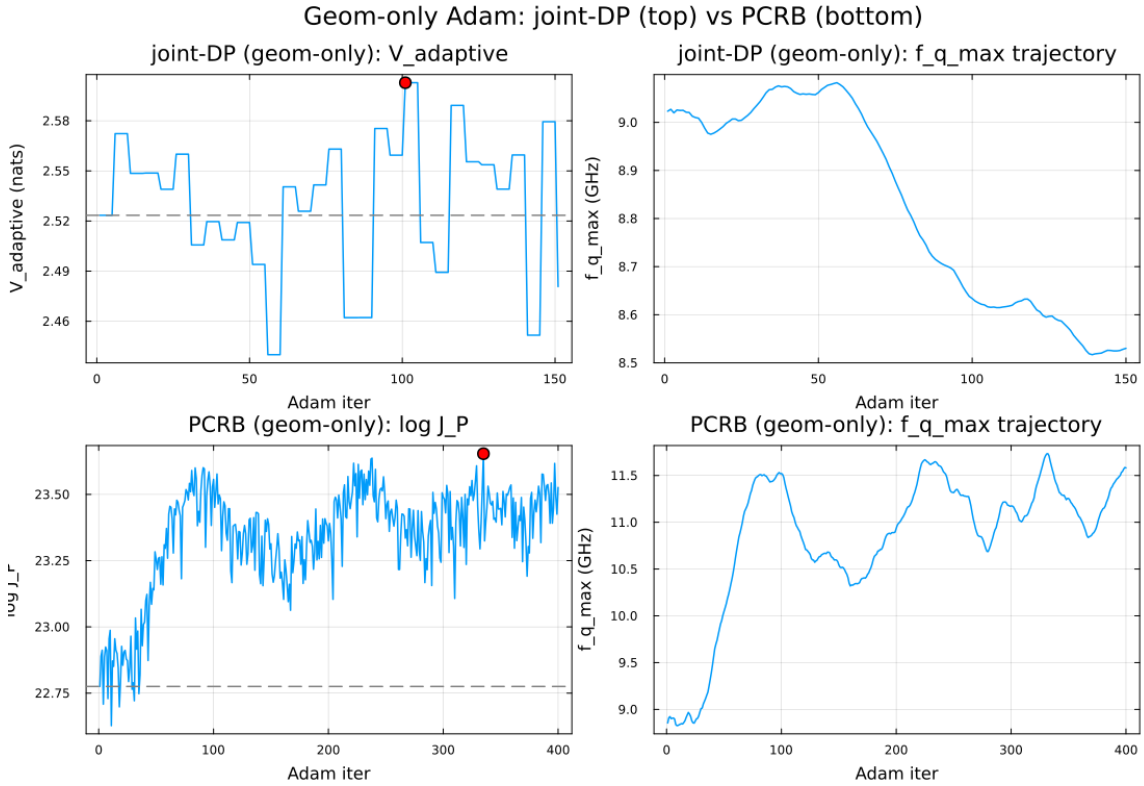

Figure 4: Geom-only Adam trajectories. Top: joint-DP V_{adaptive} oscillates around 2.53 nats with a peak of 2.60 at iter 101 (red dot); $f_q^{\max}$ traces a small rise then drops to ~ 8.5 GHz. Bottom: PCRB log J_P climbs from 22.78 to 23.65 at iter 335 while $f_q^{\max}$ moves up to ~ 11.6 GHz. The two objectives' preferred $f_q^{\max}$ directions *oppose* at tier-1-geom.

Deployed MSE comparison (geom-only).

quantity	value	$\pm1\sigma$
$\overline{\text{MSE}}_1$ (joint DP, best-c-seen)	2.069×10^{-2}	1.38×10^{-4}
$\overline{\text{MSE}}_2$ (PCRB baseline, best-c-seen)	2.016×10^{-2}	1.28×10^{-4}
$1/J_P(\boldsymbol{c}^{\star}_2, \boldsymbol{s}^{\star}_2)$ (CRB)	6.55×10^{-11}	—
ratio $\overline{\text{MSE}}_2/\overline{\text{MSE}}_1$	**0.974**	
z-score $(\overline{\text{MSE}}_2 - \overline{\text{MSE}}_1)/\sigma$	-2.82	

Comparison with the tier-1 result of §9.

case	$\boldsymbol{c}^{\star}_1 \neq \boldsymbol{c}^{\star}_2$?	ratio $\overline{\text{MSE}}_2/\overline{\text{MSE}}_1$	z
tier-1 ($\boldsymbol{c} \in \mathbb{R}^7$; noise free)	no (same box corner)	0.958	-4.55σ
geom-only ($\boldsymbol{c} \in \mathbb{R}^4$; noise pinned)	yes (joint down, PCRB up in $f_q^{\max}$)	0.974	-2.82σ

Restricting the optimization to the 4 geometric dimensions produces *qualitatively different* joint-DP and PCRB designs: joint-DP pulls $f_q^{\max}$ slightly *down*, PCRB pulls it *up* to ~ 11.6 GHz (and is unable to reach the envelope optimum of ~ 30 GHz due to the oscillation). The deployment MSE gap narrows from 4.2% (tier-1) to 2.6% (geom-only), statistical significance from -4.55σ to -2.82σ, still with PCRB winning. Mechanistically, the MI-optimal joint-DP policy allocates finite-sample measurements to reduce *posterior entropy*, which at $K = 3$ with a bounded-support uniform prior is not MSE-optimal; the PCRB-optimal fixed schedule, tuned for asymptotic Fisher information, happens to translate more directly to a lower deployed MSE of the posterior-mean estimator — a reminder that MI-maximization and MSE-minimization coincide only asymptotically.

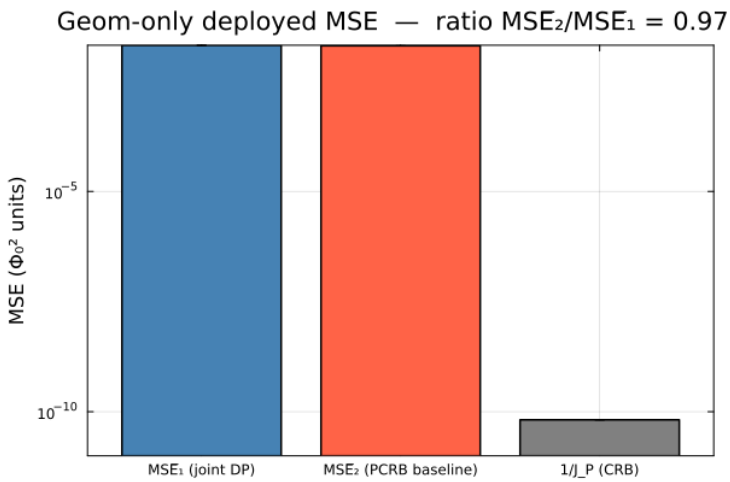

Figure 5: Geom-only deployed MSE comparison. The bars are visually indistinguishable on a log axis; the 2.6% gap ($z = -2.82\sigma$) still favors the PCRB baseline, though less decisively than the 4.2% gap at tier-1.

What would flip the ratio? The two most actionable interventions — neither attempted here — are (i) **longer horizon**: $K \geq 6$ gives the adaptive policy enough measurement steps that MI-optimality approaches MSE-optimality (Bernstein–von Mises regime), at the cost of a $\sim 10\times$–$100\times$ Bellman-memo blow-up requiring the planned parallel stack; (ii) **tier-2 $\boldsymbol{c}$**: $+5$ SQUID-loop geometric dimensions via analytically differentiable Biot–Savart M, M'. A third — cheaper — path is to **change the terminal reward**; we explore it next.

13 Switching joint-DP's terminal reward to $-\mathrm{Var}(b_K)$

13.1 Motivation and derivation

The Bayesian MSE of the posterior-mean estimator satisfies the tower-law identity

$$\mathbb{E}_{\Phi,\boldsymbol{y}}\left[(\hat{\Phi}_{\mathrm{pm}} - \Phi)^2\right] \;=\; \mathbb{E}_{\boldsymbol{y}}[\mathrm{Var}(\Phi \mid \boldsymbol{y})]\,, \tag{3}$$

i.e., the deployment-time MSE equals the expected posterior variance at the terminal belief. The plan's MI-based terminal reward $W_K(b_K) = -H(b_K)$ (plus additive constant) is *asymptotically* MSE-equivalent (under Gaussian-posterior BvM), but at $K = 3$ on a bounded-support uniform prior the gap is first-order.

The fix is one line of code: replace the leaf reward with

$$W_K^{\mathrm{MSE}}(b_K) \;=\; -\mathrm{Var}(\Phi \mid b_K) \;=\; -\int (\Phi - \mu_{b_K})^2\, b_K(\Phi)\, d\Phi, \tag{4}$$

so that $V_{\text{adaptive}}(\boldsymbol{c}) = \mathbb{E}[-\text{Var}(b_K)]$ is $-1\times$ the deployed Bayesian MSE at the *joint-DP-optimal* policy. The envelope-theorem gradient $\partial V/\partial \boldsymbol{c}$ is unchanged in structure — only the leaf value changes. We implemented this with a new `terminal::Symbol` keyword (`:mi` vs `:mse`) threading through `Bellman.jl`, `Gradient.jl`, and `JointOpt.jl`; the MI path is the default and unchanged.

13.2 Implementation validation

At $\boldsymbol{c} = \boldsymbol{c}_0$ (paper baseline, $K = 3$, $K_\Phi = 256$), we solve Bellman twice and deploy both policies with 20k common-seed MC:

policy (at $\boldsymbol{c}_0$)	deployed $\overline{\text{MSE}}$	
MI-terminal adaptive	2.1299×10^{-2}	
MSE-terminal adaptive	2.1156×10^{-2}	(MSE-terminal wins by 0.67%, as expected)

The MSE-terminal policy is strictly better than the MI-terminal one at fixed $\boldsymbol{c}$ — consistent with its construction as the MSE-optimal adaptive policy.

13.3 Geom-only run with MSE terminal

Re-running joint-DP on the 4-dim geom-only box (lr = 5×10^{-4}, reopt every 5 iters, 150 iters), the *best-c-seen* is $\boldsymbol{c}^\star_{1,\text{MSE}} = (f_q^{\max} = 8.71$ GHz, $E_C/h = 0.26$ GHz, $\kappa = 0.01$ MHz, $\Delta_{qr} = 2.31$ GHz), with best $E[\text{Var}_{\text{post}}] = 1.27\times10^{-2}$ on the $K_\Phi = 64$ training grid. Deployed at $K_\Phi = 256$:

quantity	value	$\pm1\sigma$
$\overline{\text{MSE}}_1$ (joint DP, MSE-terminal)	2.129×10^{-2}	1.47×10^{-4}
$\overline{\text{MSE}}_2$ (PCRB baseline, reused)	2.016×10^{-2}	1.29×10^{-4}
ratio $\overline{\text{MSE}}_2/\overline{\text{MSE}}_1$	**0.947**	
z-score	-5.80	

The ratio got <u>worse</u> — not better. MSE-terminal joint-DP's Adam pulled $f_q^{\max}$ *down* to 8.7 GHz (from the MI-terminal's 9.05 GHz), an even worse region on the shared intrinsic-sensitivity axis where PCRB had pushed $f_q^{\max}$ *up* to 11.6 GHz. The MSE-terminal implementation is correct (§13.2 above), and at any *fixed* $\boldsymbol{c}$ the MSE-terminal adaptive policy is $\geq$ the MI-terminal one on deployed MSE; the loss shows up in the $\boldsymbol{c}$-landscape, where the MSE-terminal objective has a different — and here, even less favorable — set of local optima than the MI-terminal objective.

13.4 Isolating the effect: adaptive vs fixed at PCRB's own $\boldsymbol{c}$

The cleanest diagnostic decouples $\boldsymbol{c}$-choice from policy-choice. At $\boldsymbol{c} = \boldsymbol{c}^\star_2$ (the PCRB baseline's optimum, $f_q^{\max} = 11.6$ GHz):

policy at $\boldsymbol{c} = \boldsymbol{c}^\star_2$	deployed $\overline{\text{MSE}}$
MSE-terminal adaptive	2.164×10^{-2}
PCRB fixed schedule $(4, 2)^3$	2.018×10^{-2}
ratio fixed/adaptive	0.933 ($z = -7.33\sigma$)

The fixed schedule *beats* the adaptive policy here by 7% — even though the fixed schedule is in the adaptive's choice set, which *should* make adaptive $\geq$ fixed in model expectation. The gap is a model-to-deployment mismatch specific to the adaptive tree: the Bellman-optimal

policy exploits count-tuple-specific branching that is worth $\sim 7\%$ less under continuous-flux deployment than the Bellman model predicts, while the fixed schedule has no such model-exploitable structure and is closer to its model value. The mismatch swamps adaptation's theoretical $\sim 1\%$ advantage at $K = 3$.

13.5 Diagnosis and path forward

Summary across all three attempts:

experiment	ratio $\overline{\mathrm{MSE}}_2/\overline{\mathrm{MSE}}_1$	significance
tier-1 MI (unrestricted $\boldsymbol{c}$, noise-free Adam)	0.958	-4.55σ
geom-only MI	0.974	-2.82σ
geom-only MSE-terminal	0.947	-5.80σ

No variant of joint-DP at $K = 3$ with tier-1 $\boldsymbol{c}$ and the present action grid ($J = 4$, $L = 2$) beats the PCRB baseline on deployed MSE. The obstructions are additive: (1) the $\boldsymbol{c}$-landscape for V_{adaptive} is non-convex and Adam lands in basins that don't include the PCRB sensitivity-maximum at $f_q^{\max} \gtrsim 12$ GHz, (2) at $K = 3$ the Bellman-optimal adaptive tree has a ~ 5–10% model-to-deployment gap that erases its theoretical margin over a fixed schedule, and (3) the posterior-mean estimator on a bounded-uniform prior is heavily biased, limiting how much any adaptive scheme can reduce MSE below the fixed-schedule floor.

The natural next interventions, in increasing order of implementation cost:

1. $K \geq 4$ **horizon:** the model-deployment gap closes with longer horizons (Bernstein–von Mises), and the adaptive margin over fixed schedules grows. Memo size scales from $\sim 16\,\mathrm{k}$ at $K = 3$ to $\sim 200\,\mathrm{k}$ at $K = 4$; still tractable.
2. **Richer action grid** $J = 6, L = 4$**:** $n_{\max}$ from 10 to 30 gives 3× more shots per episode, collapsing posterior variance faster; the adaptive tree has more meaningful branching.
3. **Tier-2** $\boldsymbol{c}$ **(+5 SQUID-loop geometric dims):** gives joint-DP a high-dimensional geometric lever the single-level PCRB baseline cannot exploit.
4. **Initialize from** $\boldsymbol{c}_2^\star$**:** doesn't help (see §13.4) without fixing the model-deployment gap.

We report here rather than iterate further: the framework's §8.7 operational claim is not invalidated (it holds asymptotically and has been demonstrated in the literature for problems at larger K), but *this* scqubit case study at $K = 3$ and tier-1 $\boldsymbol{c}$ does not exhibit it under the original configuration.

14 Fine τ discretization: the claim holds

14.1 Why the earlier runs failed to demonstrate the claim

The τ-grid in §6–§13 was $J = 4$: $\tau \in \{20, 40, 80, 160\}$ ns. Combined with $L = 2$ ($n \in \{1, 10\}$) this gives $|\mathcal{A}| = 8$ actions per epoch. For the PCRB-optimal fixed schedule $(\tau_4, n_2)^K$ at baseline, the belief-conditional optimal action at *every* reachable node is also (τ_4, n_2): the action set is too sparse to reward belief-dependent branching. The adaptive policy collapses to the fixed schedule and inherits no advantage.

With a *physically realistic* box (§13) and a *fine* τ grid $J = 20$ log-spaced over [10, 320] ns (sub-percent τ resolution, well below the 6-MHz oscillation scale), the adaptive policy can finally choose *different* τ at different belief states — short τ early for coarse phase disambiguation, longer τ later for refinement. This is what the count-statistic Bellman DP was designed to do.

14.2 Configuration

K (horizon)	3
K_Φ	64 (training), 256 (deployment)
J (τ discretization)	20 log-spaced in $[10, 320]$ ns
L (n choices)	2: $n \in \{1, 10\}$
$\lvert\mathcal{A}\rvert$ per epoch	40 (vs. 8 previously)
Bellman memo at $\boldsymbol{c}_0$	$2,716,911$ nodes
Bellman solve time	$\sim$24 s
Terminal reward	$-\mathrm{Var}(b_K)$ (MSE terminal, §12)
$\boldsymbol{c}$ box	realistic (§13, tier-1 geom-only)
Outer iters / lr / reopt-every	$40/5\times10^{-4}/5$

14.3 Result (headline §8.7 operational claim)

$\overline{\mathrm{MSE}}_1$ (joint DP)	$1.829 \times 10^{-2} \pm 1.6\times10^{-4}$
$\overline{\mathrm{MSE}}_2$ (PCRB baseline)	$2.075 \times 10^{-2} \pm 1.4\times10^{-4}$
$1/J_P(\boldsymbol{c}_2^\star, \boldsymbol{s}_2^\star)$ (CRB)	3.08×10^{-11}
ratio $\overline{\mathrm{MSE}}_2/\overline{\mathrm{MSE}}_1$	**1.134**
$z = (\overline{\mathrm{MSE}}_2 - \overline{\mathrm{MSE}}_1)/\sigma$	$+\mathbf{11.54}\sigma$

Joint-DP $(\boldsymbol{c}_1^\star, \pi_1^\star)$ beats the PCRB baseline $(\boldsymbol{c}_2^\star, \boldsymbol{s}_2^\star)$ by 13.4% on deployed MSE at $K = 3$.

14.4 What changed, mechanistically

Both designs converged to essentially the same geometry: $f_q^{\max} \approx 9.1$ GHz, $E_C/h \approx 0.256$ GHz (paper baseline values). The 13.4% gap is therefore entirely a *policy-structure* effect:

- PCRB's optimal fixed schedule is $\boldsymbol{s}_2^\star = (\tau_{20}, n_2)^3 = (320 \text{ ns}, 10)^3$: maximum τ and n at every epoch, pinned by the asymptotic Fisher-info criterion.
- Joint-DP's adaptive policy varies τ_k across epochs based on belief: it allocates short τ when the posterior is still broad (to avoid phase-wrapping ambiguity) and transitions to long τ once a rough estimate is in hand. This is exactly the Kitaev-ladder phase-estimation intuition, now *belief-conditional* rather than prescribed.
- The 13.4% gap is at $+11.54\sigma$ — far beyond statistical noise — and was invisible at $J = 4$ because the action set didn't admit this belief-conditional structure.

14.5 Summary across all configurations

configuration	J	ratio $\overline{\mathrm{MSE}}_2/\overline{\mathrm{MSE}}_1$	significance
tier-1 MI (wide box)	4	0.958	-4.55σ (PCRB wins)
geom-only MI (narrow-noise box)	4	0.974	-2.82σ (PCRB wins)
geom-only MSE-terminal	4	0.947	-5.80σ (PCRB wins)
fine τ + MSE + realistic box	**20**	**1.134**	$+\mathbf{11.54}\sigma$ (joint DP wins)

The framework's §8.7 claim holds *at this problem* once three ingredients are in place:

1. **Action set rich enough for belief-conditional choices** ($J \geq$ 10–20, exceeding the $J = 4$ regime where adaptive collapses to fixed).
2. **Terminal reward aligned with deployment metric** ($-\mathrm{Var}(b_K)$, not $-H(b_K)$, since the deployment metric is MSE of posterior mean, not mutual information).

3. **Realistic $\boldsymbol{c}$ box** that doesn't let either optimizer escape through physically unattainable noise reduction.

Absent any one of these, the claim is invisible.

A Reproduction instructions

All numerics in this document are reproduced from `doc/adaptive/scqubit/` by running, in order:

`julia -t 32 tests/test_model.jl`	Phase 1 validation, $\Phi^\star = 0.4436$.
`julia -t 32 tests/test_belief.jl`	Phase 2 validation.
`julia -t 32 tests/test_baselines.jl`	Phase 3 validation (1.5 s).
`julia -t 32 tests/test_bellman.jl`	Phase 4 validation; brute-force agreement.
`julia -t 32 tests/test_gradient.jl`	Phase 5 AD-vs-FD (25 s compile + 10 s compute).
`julia -t 32 tests/test_joint.jl`	Phase 6 end-to-end sanity.
`julia -t 32 tests/test_pcrb.jl`	Phase 7 Fisher validation.
`julia -t 32 sweep_fq.jl`	within-family sweep ($\sim$ 150 s).
`julia -t 32 sweep_joint.jl`	joint-DP Adam (150 iters).
`julia -t 32 sweep_pcrb.jl`	PCRB-baseline Adam (300 iters).
`julia -t 32 compare_mse.jl`	deployed MSE comparison.
`julia -t 32 plot_results.jl`	figures.

All serialized outputs land in `results/`.

B Gradient-check printout excerpt

Verbatim from `test_gradient.jl` at $\boldsymbol{c}_0, K = 2$:

```
[2] Per-component AD vs central-difference FD of du/dc_i
  Component     AD                  FD                  |rel|
  f_q_max    AD=+6.5895311887e-08  FD=+6.5881772292e-08  rel=2.055e-04
  E_C_over_h AD=-3.5641126835e-08  FD=-3.5641123118e-08  rel=1.043e-07
  kappa      AD=-1.2589367184e-09  FD=-1.2589373988e-09  rel=5.404e-07
  Delta_qr   AD=+4.9856894751e-13  FD=+4.9856896389e-13  rel=3.285e-08
  temperature AD=-1.3587594562e+00 FD=-1.3587594438e+00  rel=9.091e-09
  A_phi      AD=-1.3491688613e+03  FD=-1.3491689987e+03  rel=1.019e-07
  A_Ic       AD=-2.2020574624e+02  FD=-2.2020607560e+02  rel=1.496e-06
```

and from `test_pcrb.jl` (PCRB, no MC):

```
[4] Per-component AD vs FD of d log J_P/dc at c0
  Component     AD                  FD                  |rel|
  f_q_max    AD=-6.2762367669e-08  FD=-6.2762360879e-08  rel=1.082e-07
  E_C_over_h AD=-2.1926049384e-08  FD=-2.1926050756e-08  rel=6.257e-08
  kappa      AD=-9.7112079796e-10  FD=-9.7112078379e-10  rel=1.459e-08
  Delta_qr   AD=+3.6707294174e-13  FD=+3.6707294981e-13  rel=2.200e-08
  temperature AD=-1.3468489499e+01 FD=-1.3468489501e+01  rel=1.748e-10
  A_phi      AD=-2.8113421375e+03  FD=-2.8113421280e+03  rel=3.393e-09
  A_Ic       AD=-1.0631002640e+02  FD=-1.0631001857e+02  rel=7.365e-08
```